

the asymptotic null distribution of  $\hat{\Lambda}$  is  $\chi_L^2$ , and the estimator is computable in linear time rather than  $\widehat{\text{MMD}}_U$ 's quadratic time.

Jitkrittum et al. (2017) jointly learn the parameters  $v_j$  and kernel parameters to optimize test power. The best such test locations ( $L = 1$ ) for a Gaussian kernel (with learned bandwidth) are shown in Figure 10. We could also try optimizing a deep kernel (1) and the test locations together; this procedure, however, failed to find a useful test. We can find a better test, though, with a two-stage scheme: first, learn a deep kernel to maximize  $\hat{J}_\lambda$ , then choose  $v_i$  to maximize  $\hat{\Lambda}$  with that kernel fixed. Results are shown in Figure 11.

Although these approaches give nontrivial test power, it is hard to interpret either set of images, as the test locations have moved far outside the set of natural images. We can instead constrain  $v_1 \in S_{\mathbb{P}} \cup S_{\mathbb{Q}}$ , simply picking the single point from the dataset which maximizes  $\hat{\Lambda}$  (shown in Figure 12). This achieves similar test power, but lets us see that the difference might lie in images with smaller objects of interest than the mean for *CIFAR-10*.

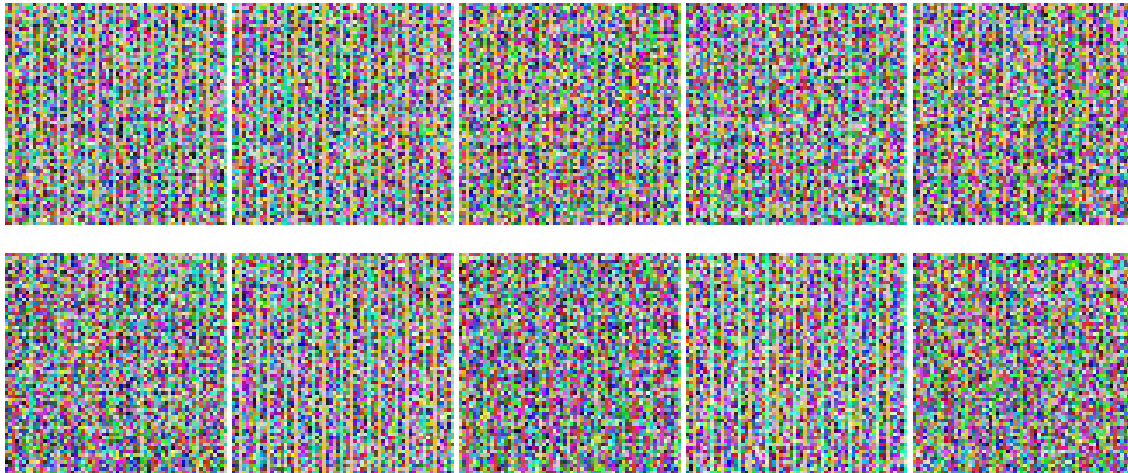


Figure 10. The best test locations (learned by an ME test with  $L = 1$ ) from 10 experiments on *CIFAR-10* vs *CIFAR-10.1*. Average rejection rate is 0.415.

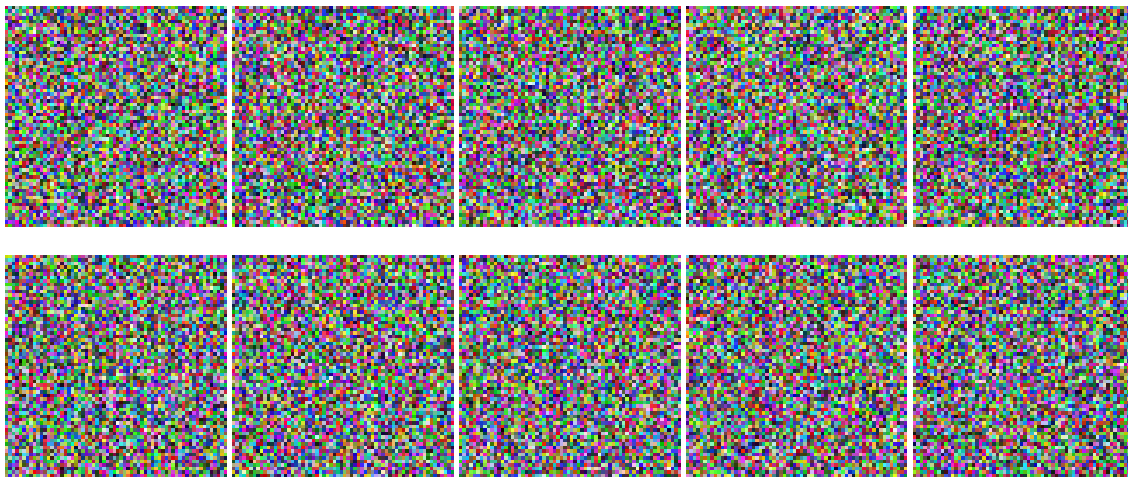


Figure 11. The best test locations (learned by an ME test,  $L = 1$ , with a deep kernel optimized for an MMD test) from 10 experiments on *CIFAR-10* vs *CIFAR-10.1*. Average rejection rate is 0.637.

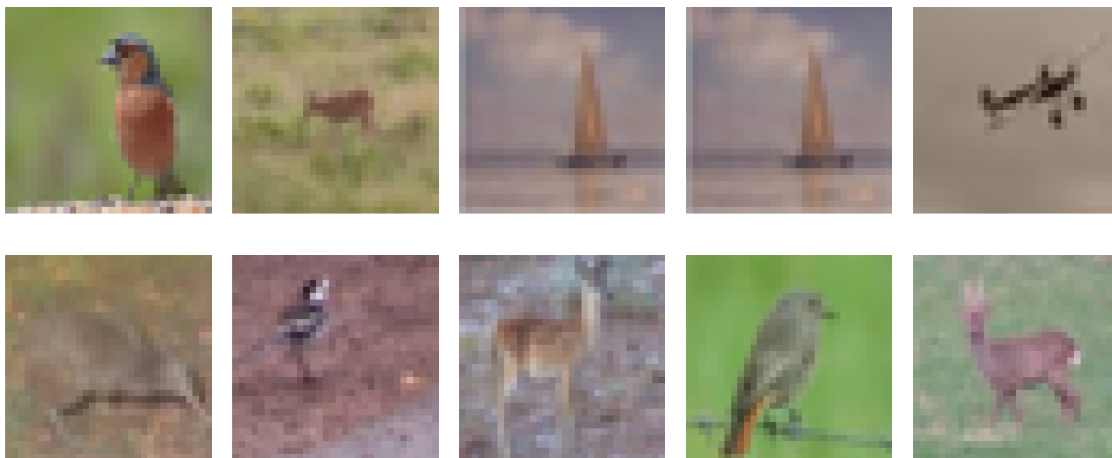


Figure 12. The best test locations (selected among existing images with our learned deep kernel,  $L = 1$ ) from 10 experiments on *CIFAR-10* vs *CIFAR-10.1*. Average rejection rate is 0.653.